COURIER MANAGEMENT SYSTEM (CMS)

FINAL PROJECT DOCUMENTATION

A Salesforce-based Courier and Logistics Management Application



Student Name	PUGAGZENTHI D
Register Number	24524U48026
Department / Class	B.Sc Data Science – III Year
Institution	Karan Arts and Science College
Academic Year	3rd Year
Team Members	Ajith E (24524U48001) Vallarasu S (24524U48034) Sriram H (24524U48030)
Guide / Faculty	NITHYA, MCA, M.Phil, B.Ed

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Screenshot evidence is embedded in the relevant project implementation, testing, reporting, and dashboard sections above.

1. INTRODUCTION

1.1 Project Overview

The Courier Management System (CMS) is a Salesforce-based application designed to digitize and centralize courier and logistics operations. The system provides an integrated environment for managing customers, shipments, delivery agents, branches, delivery updates, invoices, reports, and dashboards.

1.2 Purpose

The purpose of the Courier Management System is to automate and centralize courier and logistics operations using the Salesforce platform. The system aims to replace manual and fragmented processes with an integrated digital solution that enables efficient shipment booking, real-time tracking, delivery agent assignment, automated notifications, billing, and performance reporting. By providing a single platform for managing shipments across multiple branches, CMS improves operational visibility, customer communication, and decision-making through analytics.

1.3 Project Goals

- Centralize courier and shipment information.
- Support shipment booking and tracking.
- Manage delivery agent and branch information.
- Provide real-time delivery status updates.
- Automate shipment status changes using Salesforce Flow.
- Generate invoice information for shipments.
- Provide reports and dashboards for operational analysis.
- Apply validation rules to improve data quality.
- Support role-based access and organized Salesforce administration.

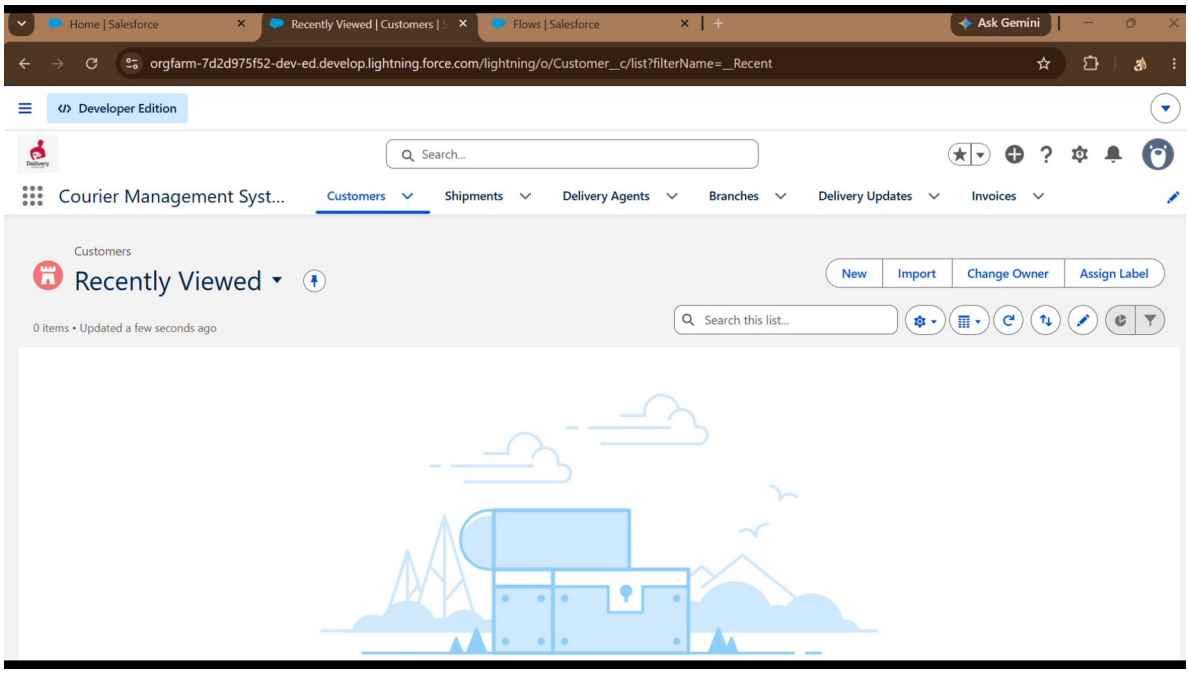


Figure 1 – Salesforce Courier Management System application/home screen

2. IDEATION PHASE

2.1 Problem Statement

The project identifies problems caused by manual shipment management and disconnected systems. Branch managers need to manage shipments efficiently when tracking is manual. Customers need timely parcel tracking updates, but delayed updates and the absence of a centralized system can create uncertainty.

2.2 Proposed Solution

The proposed solution is a centralized Salesforce Courier Management System that combines customer management, shipment booking, agent assignment, delivery updates, invoicing, reporting, and dashboards within one platform.

2.3 Business Model and Impact

- Problem: Manual shipment management.
- Solution: Salesforce-based Courier Management System.
- Innovation: Automated tracking and billing.
- Impact: Faster and more visible delivery operations.
- Business Model: Courier services.
- Scalability: Designed to support multiple branches.

3. REQUIREMENT ANALYSIS

3.1 Functional Requirements

Requirement	Description
FR-1	Customer management
FR-2	Shipment booking
FR-3	Delivery agent assignment
FR-4	Real-time status updates
FR-5	Invoice generation
FR-6	Reports and dashboards

3.2 Non-Functional Requirements

Requirement	Description
Usability	Easy Lightning interface
Security	Role-based access control
Reliability	Error-free automation
Performance	Fast tracking updates
Availability	24/7 cloud access
Scalability	Supports growing branches

3.3 Data Flow

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports

4. SYSTEM DESIGN

4.1 Custom Salesforce Objects

Object	Purpose
Customer__c	Stores customer personal and contact information.
Shipment__c	Manages parcel details, source, destination, and tracking information.

Delivery_Agent__c	Maintains delivery agent profiles and branch details.
Branch__c	Represents courier branch offices.
Delivery_Update__c	Stores status updates for each shipment.
Invoice__c	Manages billing details for shipments.

4.2 Object Relationships

- Customer → Shipment: Lookup relationship.
- Shipment → Delivery Agent: Lookup relationship.
- Delivery Agent → Branch: Lookup relationship.
- Delivery Update → Shipment: Master-Detail relationship.
- Invoice → Shipment: Lookup relationship.

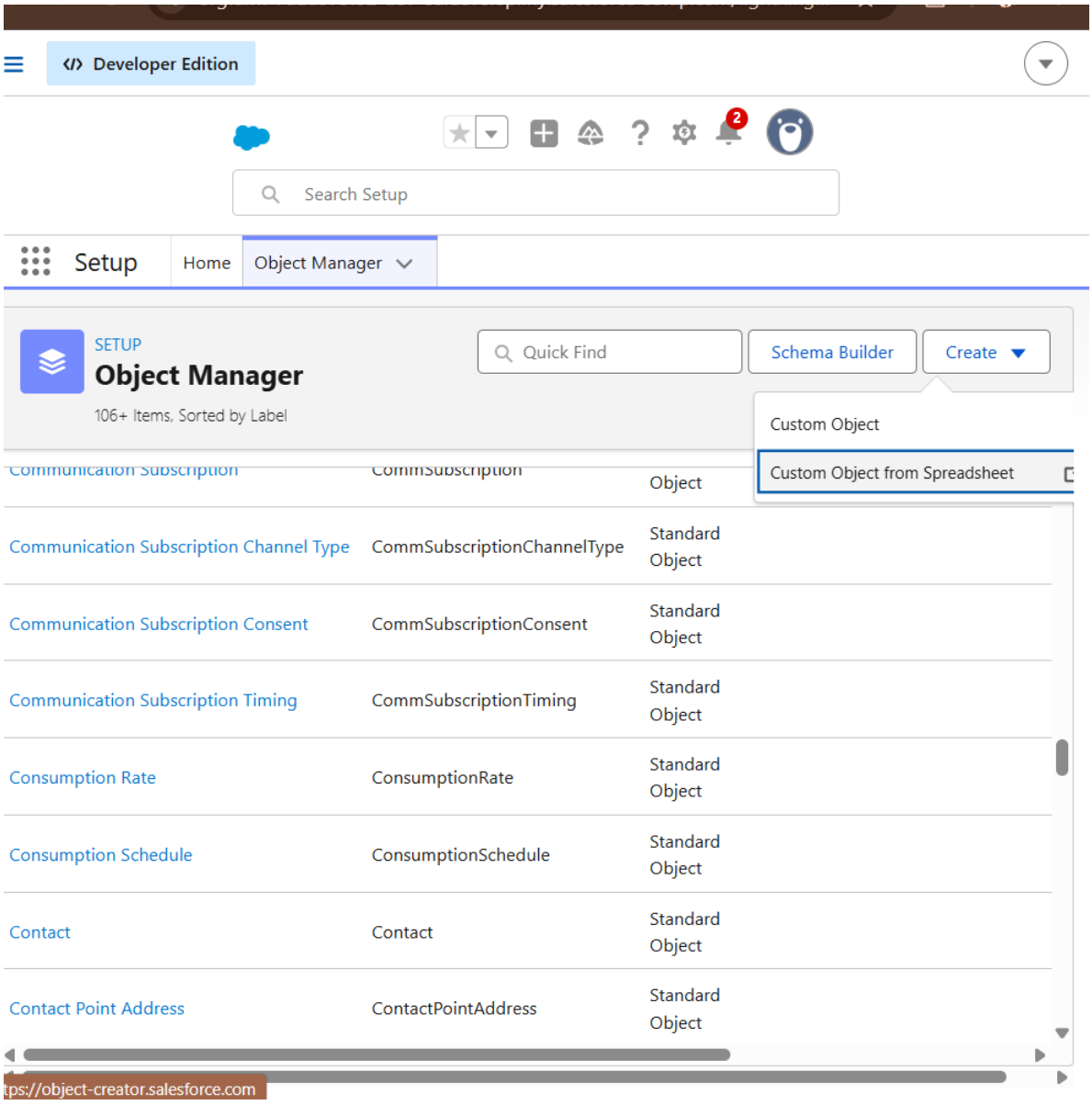


Figure 2 – Salesforce Object Manager showing the CMS custom objects

4.3 Solution Architecture

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Salesforce Flows

- Reports
- Dashboards

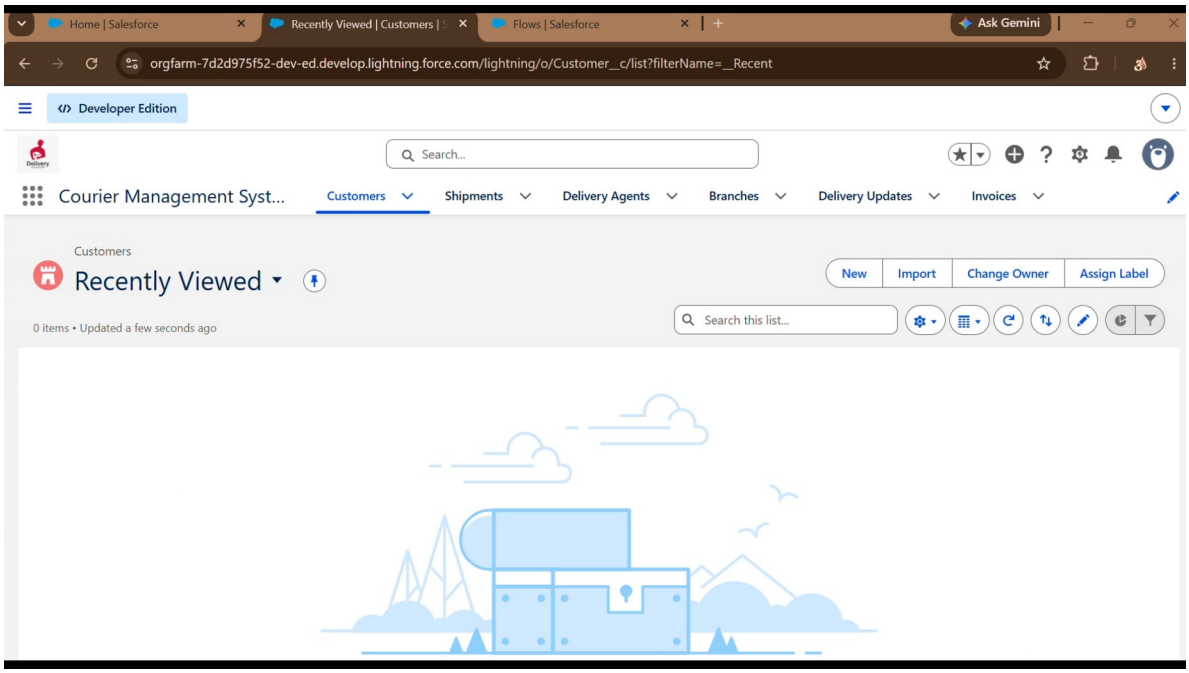
5. TECHNOLOGY STACK

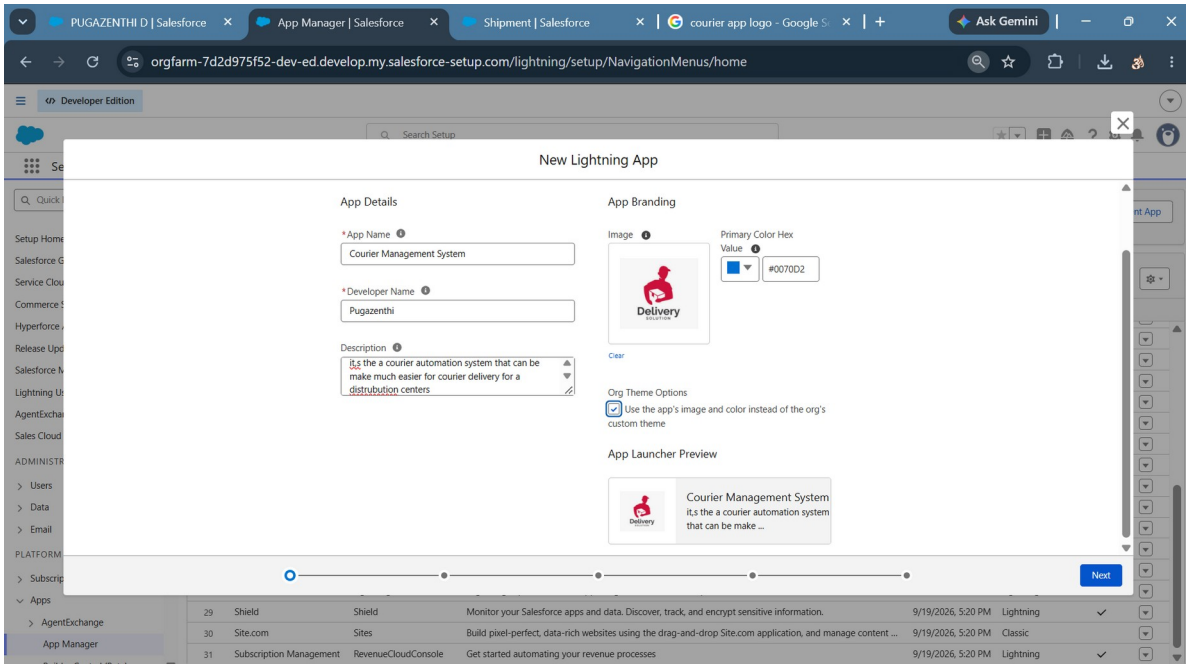
Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Flows
Security	Profiles, Roles
Reporting	Reports & Dashboards

6. PROJECT DEVELOPMENT AND SALESFORCE IMPLEMENTATION

6.1 Salesforce Account and Application

The project is implemented on Salesforce using the Lightning interface. A dedicated Courier Management System application was configured to provide access to the project objects and operational functions.


Home | Salesforce
Recently Viewed | Customers |
Flows | Salesforce
Ask Gemini
orgfarm-7d2d975f52-dev-ed.develop.lightning.force.com/lightning/o/Customer__c/list?filterName=__Recent
Developer Edition
Search...
Courier Management Syst...
Customers
Shipments
Delivery Agents
Branches
Delivery Updates
Invoices
Customers
Recently Viewed
New
Import
Change Owner
Assign Label
0 items • Updated a few seconds ago
Search this list...


PUGAZENTHI D | Salesforce
App Manager | Salesforce
Shipment | Salesforce
courier app logo - Google S
Ask Gemini
orgfarm-7d2d975f52-dev-ed.develop.my.salesforce-setup.com/lightning/setup/NavigationMenus/home
Developer Edition
New Lightning App
App Details
*App Name
Courier Management System
*Developer Name
Pugazenthi
Description
it,s the a courier automation system that can be make much easier for courier delivery for a distrubution centers
App Branding
Image
Primary Color Hex Value
#0070D2
Clear
Org Theme Options
Use the app's image and color instead of the org's custom theme
App Launcher Preview
Courier Management System
it,s the a courier automation system that can be make ...
Next

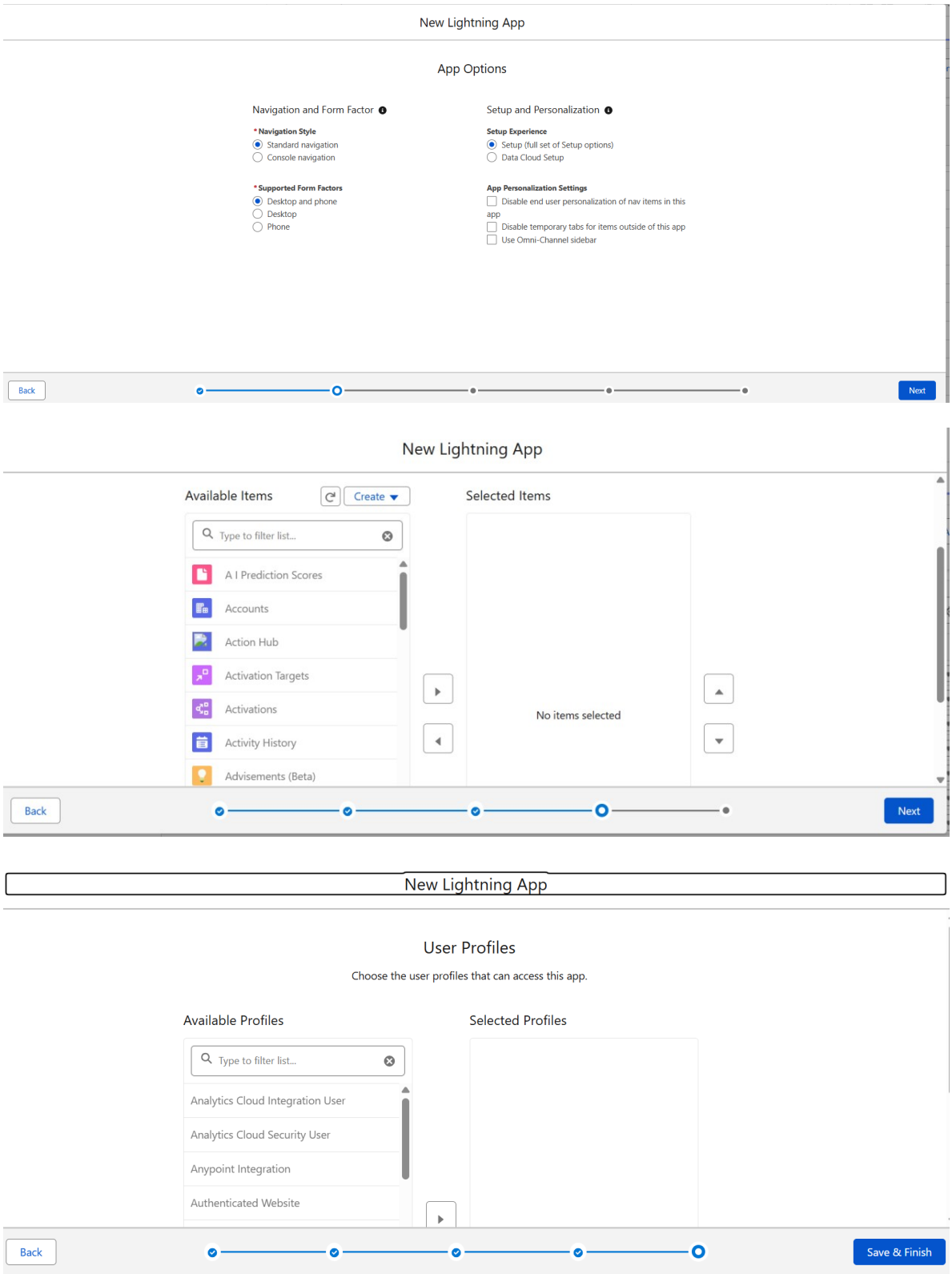


Figure 3 – Courier Management System Lightning App configuration and navigation

6.2 Custom Objects and Data Model

The CMS data model is implemented using six custom Salesforce objects. These objects organize customer, shipment, delivery, branch, update, and invoice information.

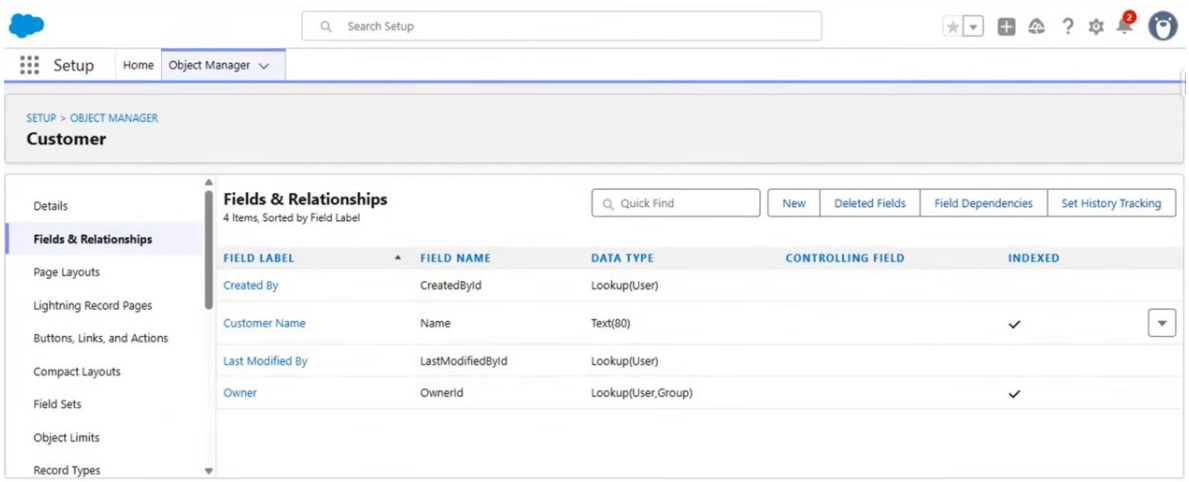


Figure 4 – Customer object fields and relationships

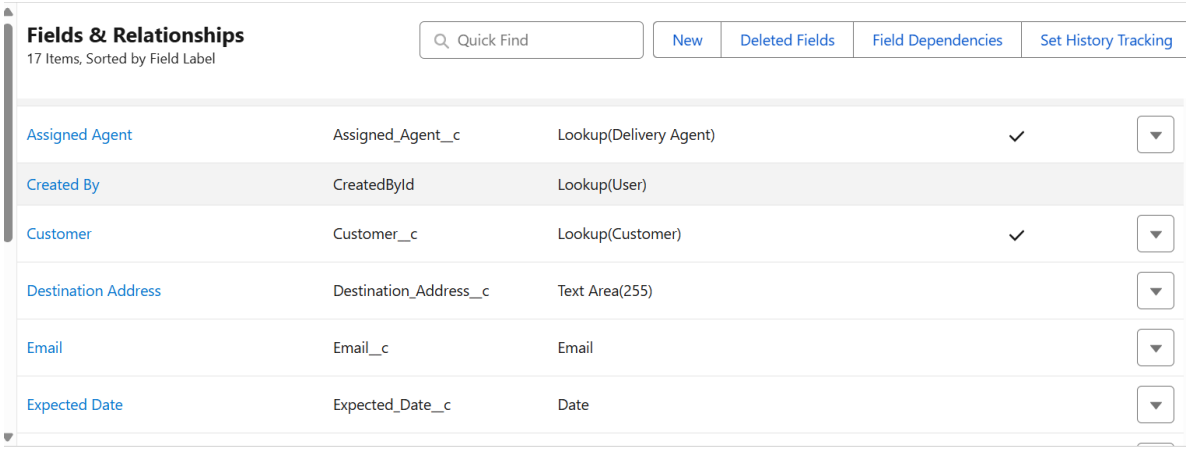


Figure 5 – Shipment object fields and configuration

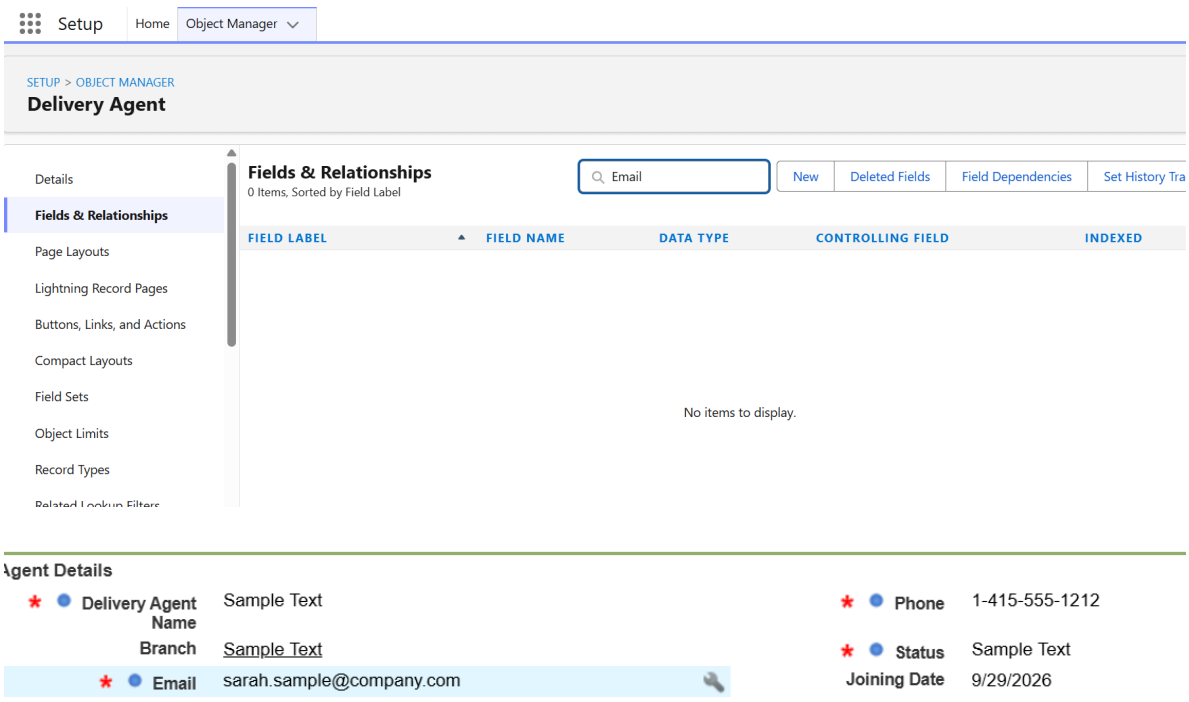


Figure 6 – Delivery Agent fields and record layout evidence

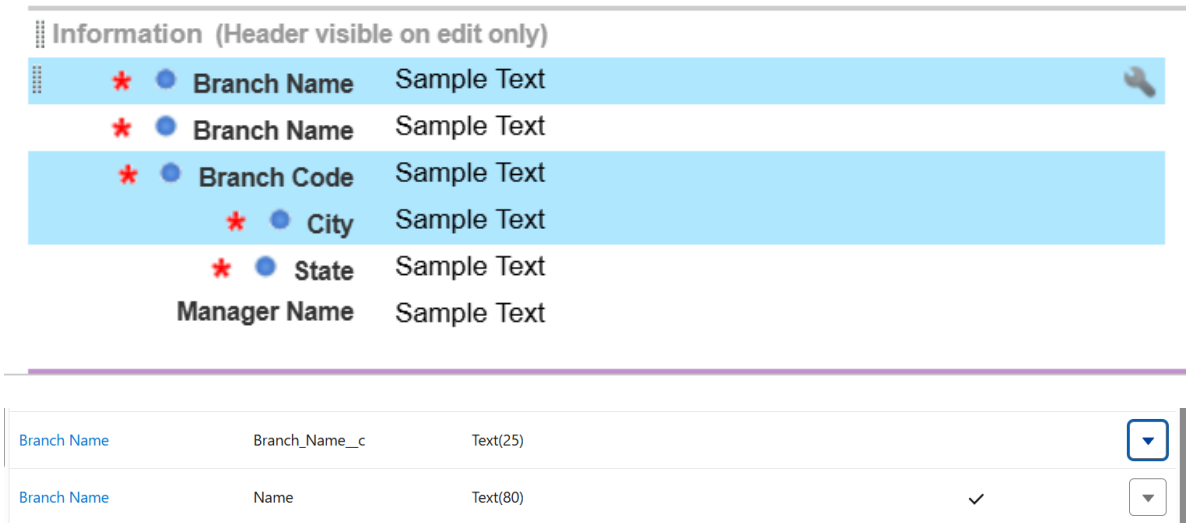


Figure 7 – Branch object fields and Branch Name configuration

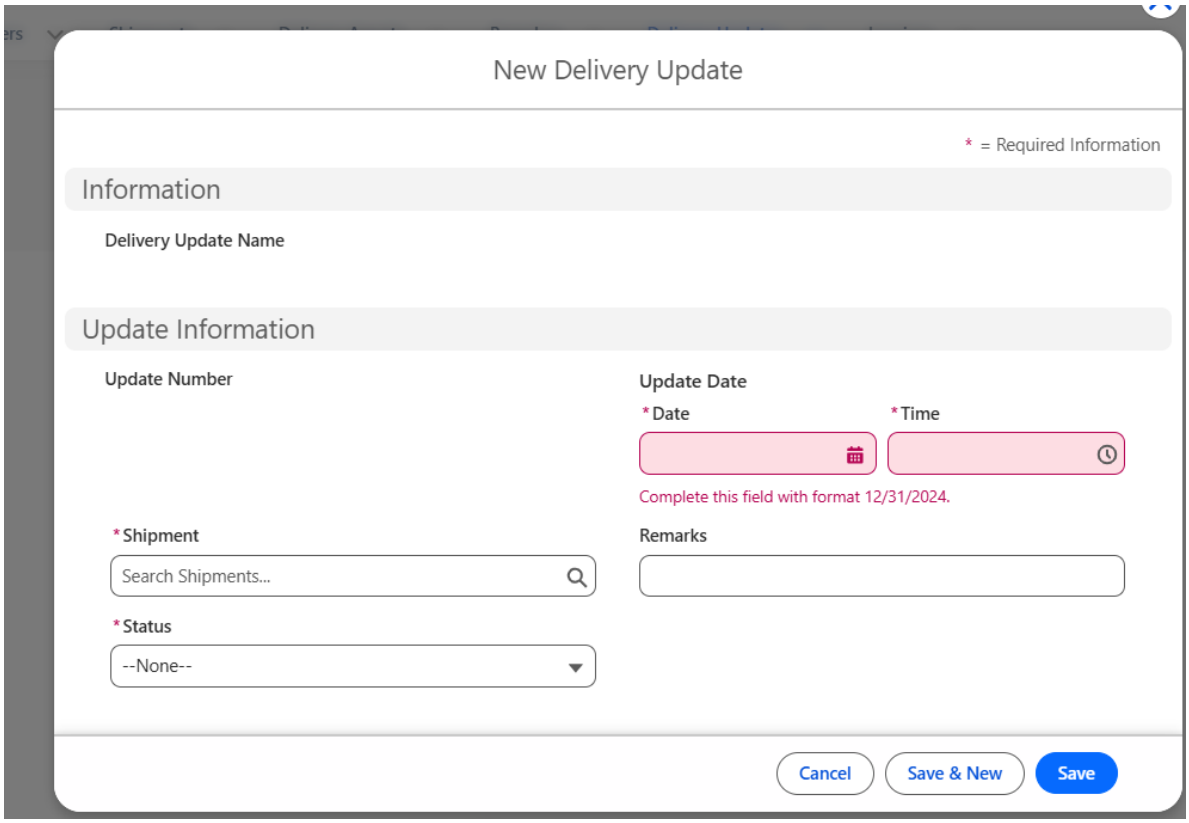


Figure 8 – Delivery Update creation and required fields

6.3 Page Layouts

Customer Layout: Customer Information, Address Details.

Shipment Layout: Shipment Details, Address Information, Delivery Information.

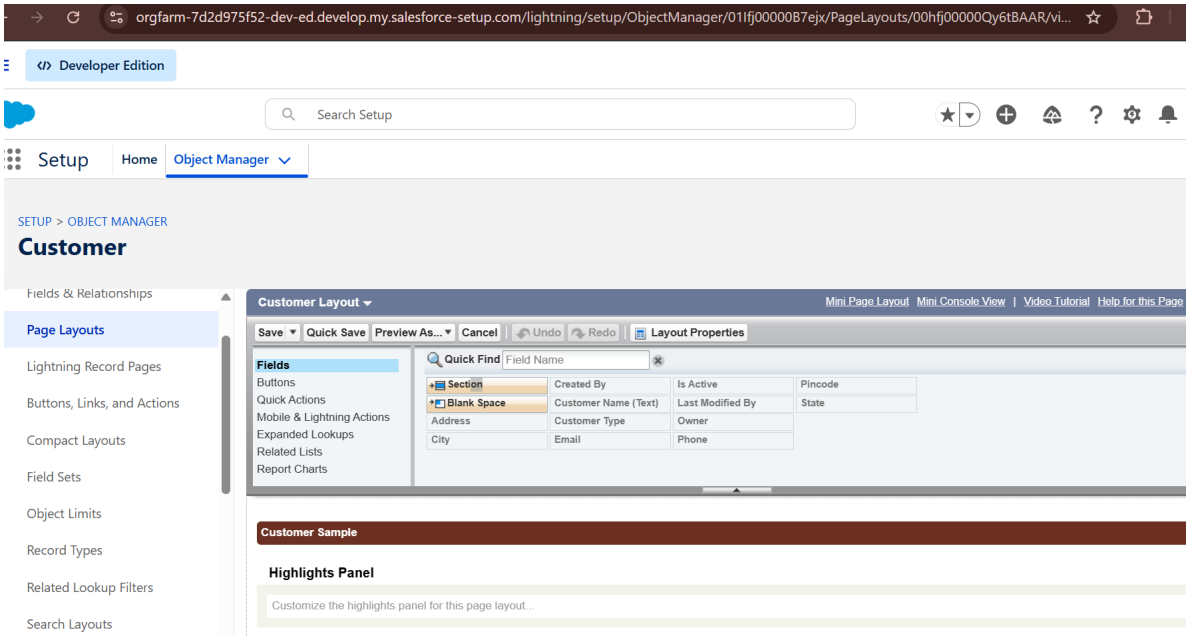
Delivery Agent Layout: Agent Details, Related Shipments.

Branch Layout: Branch Information.

Delivery Update Layout: Update Information.

Invoice Layout: Invoice Details.

Page layouts were organized to present required fields, sections, related information, and operational data in a clear Lightning record interface.



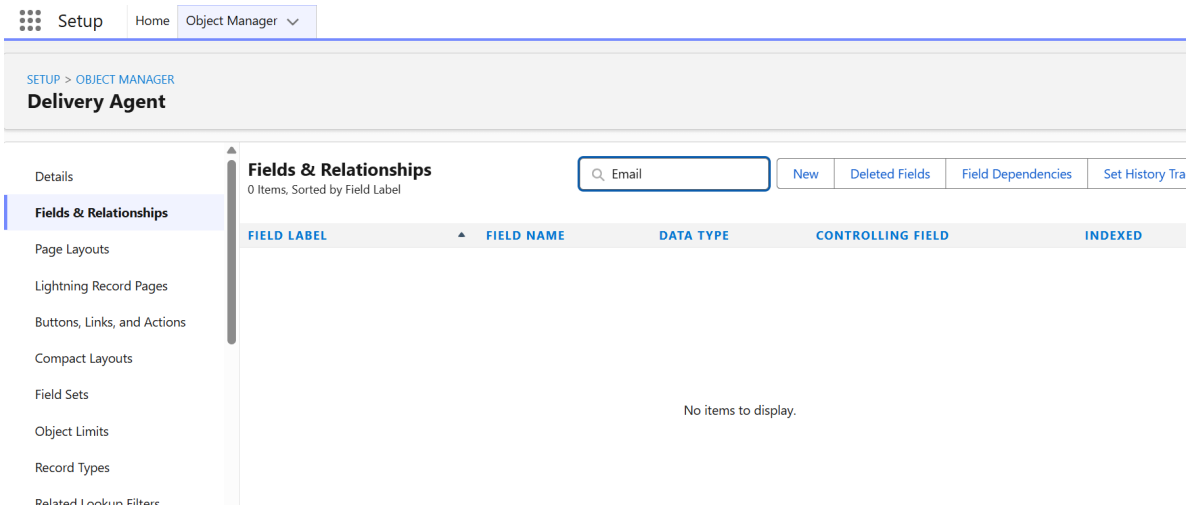
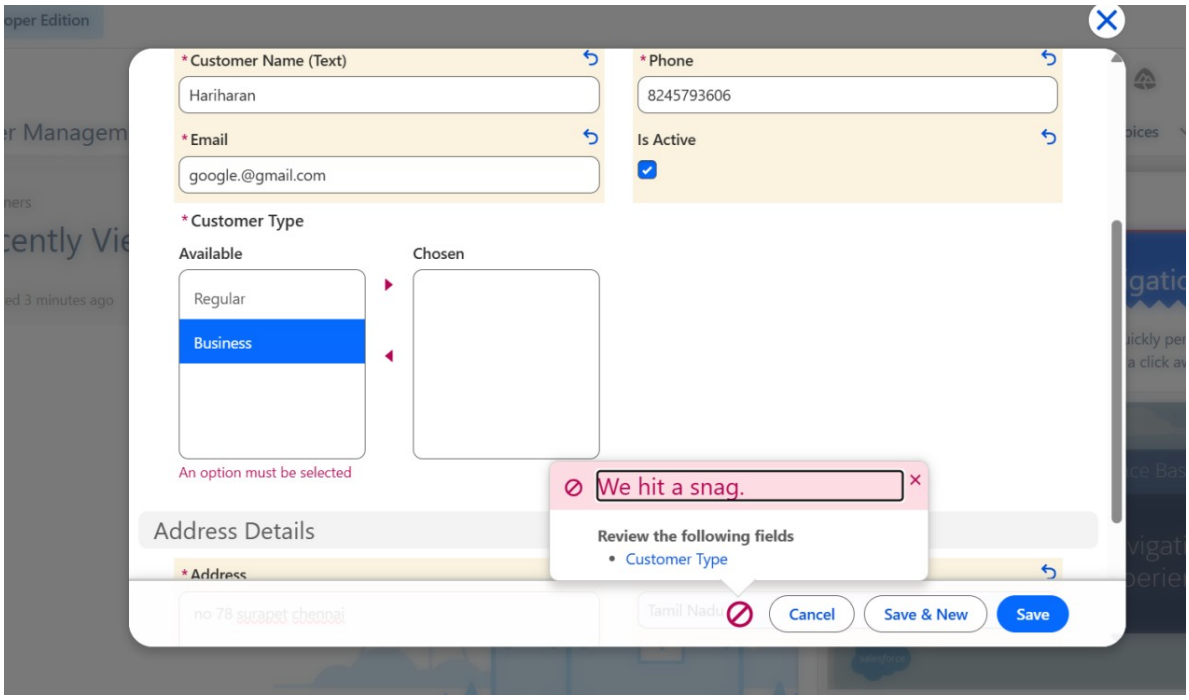


Figure 9 – Salesforce page layout configuration and record layout evidence

6.4 Validation Rules

Rule Name	Object	Condition	Error Message
Email_Required	Customer__c	ISBLANK(Email__c)	Email is required.
Validate_Phone_Number	Customer__c	LEN(Phone__c) <> 10	Phone number must be 10 digits.
Validate_Pincode	Customer__c	LEN(Pincode__c) <> 6	Pincode must be 6 digits.
Validate_Weight	Shipment__c	Weight__c <= 0	Weight must be greater than zero.
Validate_Price	Shipment__c	Price__c <= 0	Price must be greater than zero.
Validate_Expected_Date	Shipment__c	Expected_Date__c < TODAY()	Expected delivery date cannot be in the past.
Validate_Agent_Email	Delivery_Agent__c	NOT(CONTAINS(Email__c, "@"))	Enter a valid email address.
Branch_Code_Required	Branch__c	ISBLANK(Branch_Code__c)	Branch Code is mandatory.
Remarks_Required_For_Delay	Delivery_Update__c	AND(ISPICKVAL(Status__c, "Delayed"), ISBLANK(Remarks__c))	Remarks are required when status is Delayed.


*Customer Name (Text)
Hariharan
*Phone
8245793606
*Email
google.@gmail.com
Is Active
*Customer Type
Available
Chosen
Regular
Business
An option must be selected
We hit a snag.
Review the following fields
Customer Type
Address Details
*Address
Cancel
Save & New
Save

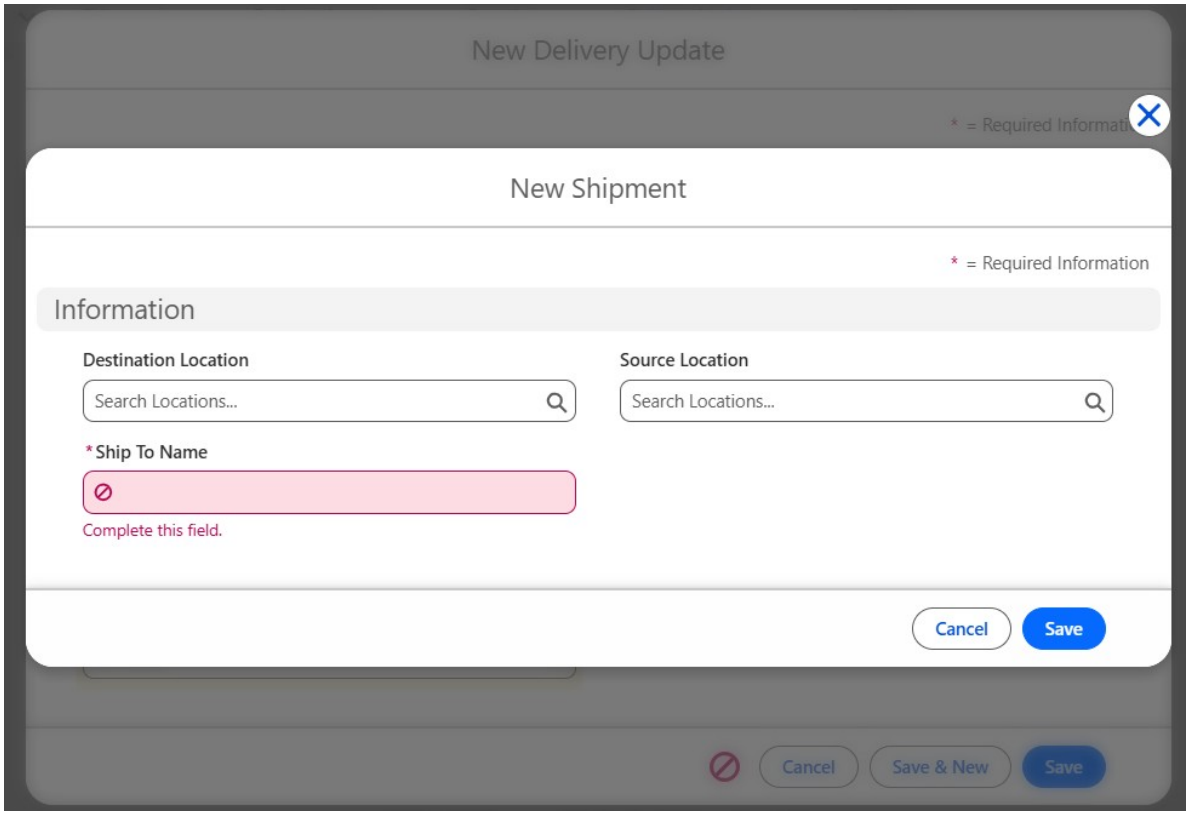
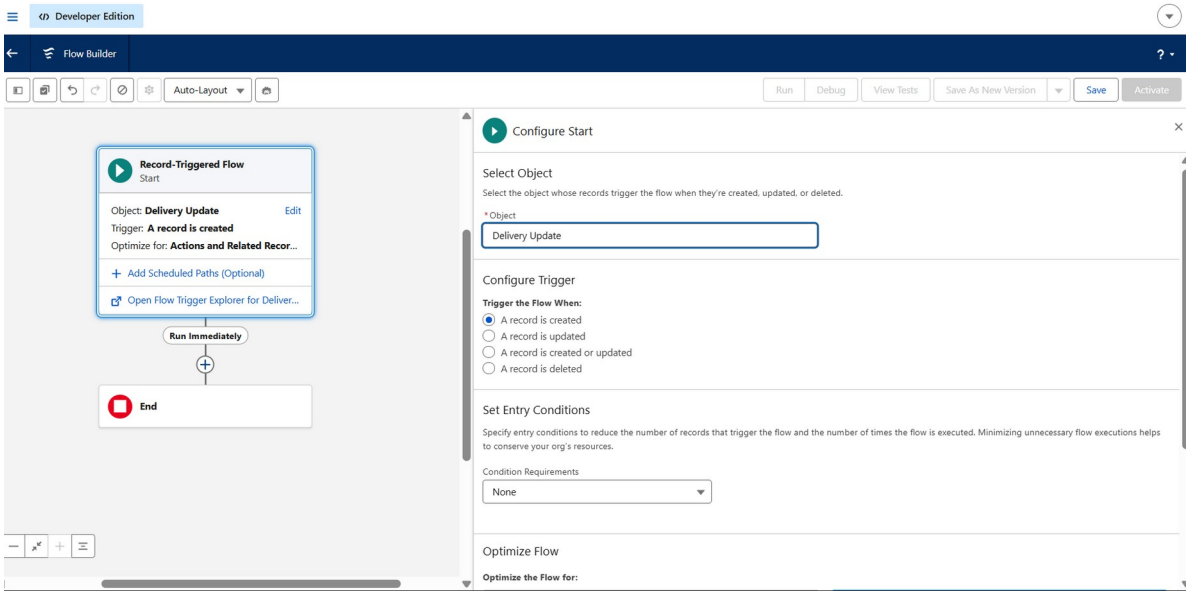

New Delivery Update
New Shipment
* = Required Information
Information
Destination Location
Search Locations...
Source Location
Search Locations...
*Ship To Name
Complete this field.
Cancel
Save
Cancel
Save & New
Save

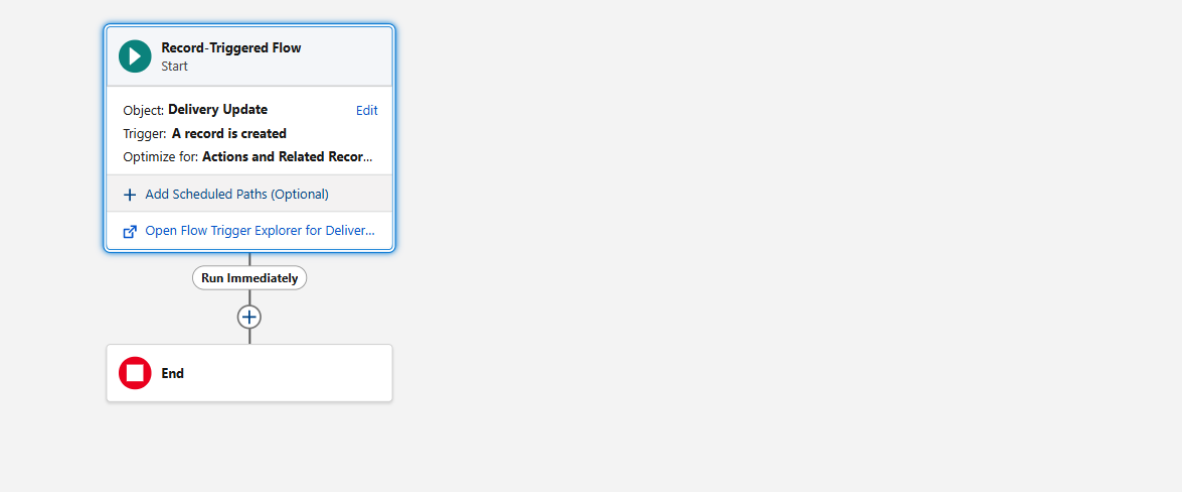
Figure 10 – Validation-rule and required-field error evidence

6.5 Automation – Delivery Status Flow

A record-triggered Salesforce Flow was implemented for Delivery_Update__c. When a delivery update is created, the related Shipment__c status is updated with the latest delivery status.

12. A Delivery_Update__c record is created.
13. The flow is triggered by the new delivery update.
14. The related Shipment__c record is identified.
15. Shipment Status__c is updated with the Delivery Update Status__c value.





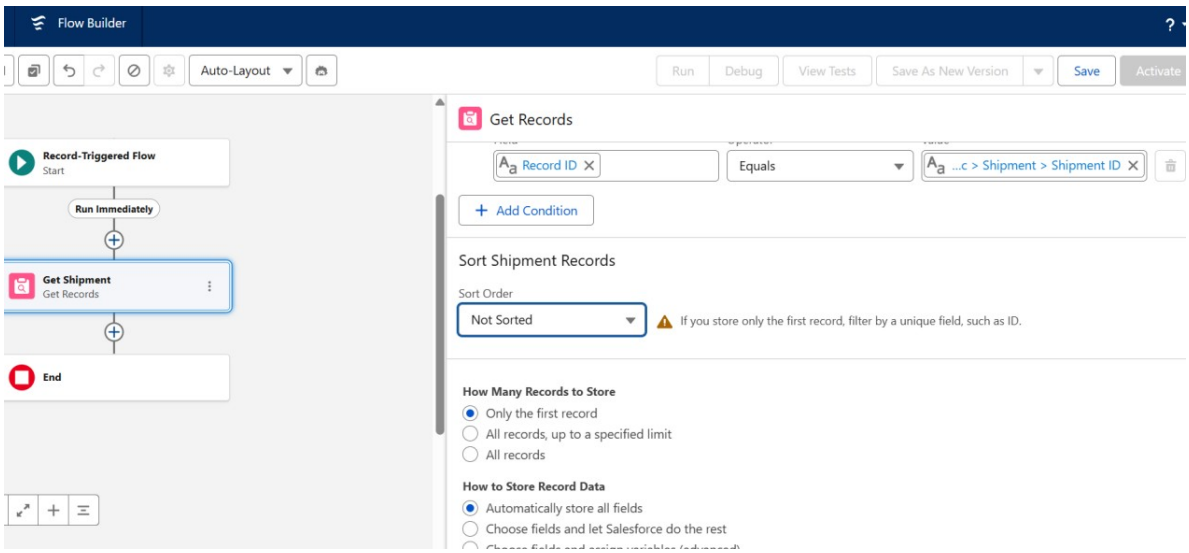


Figure 11 – Auto_Update_Shipment_Status Flow Builder configuration

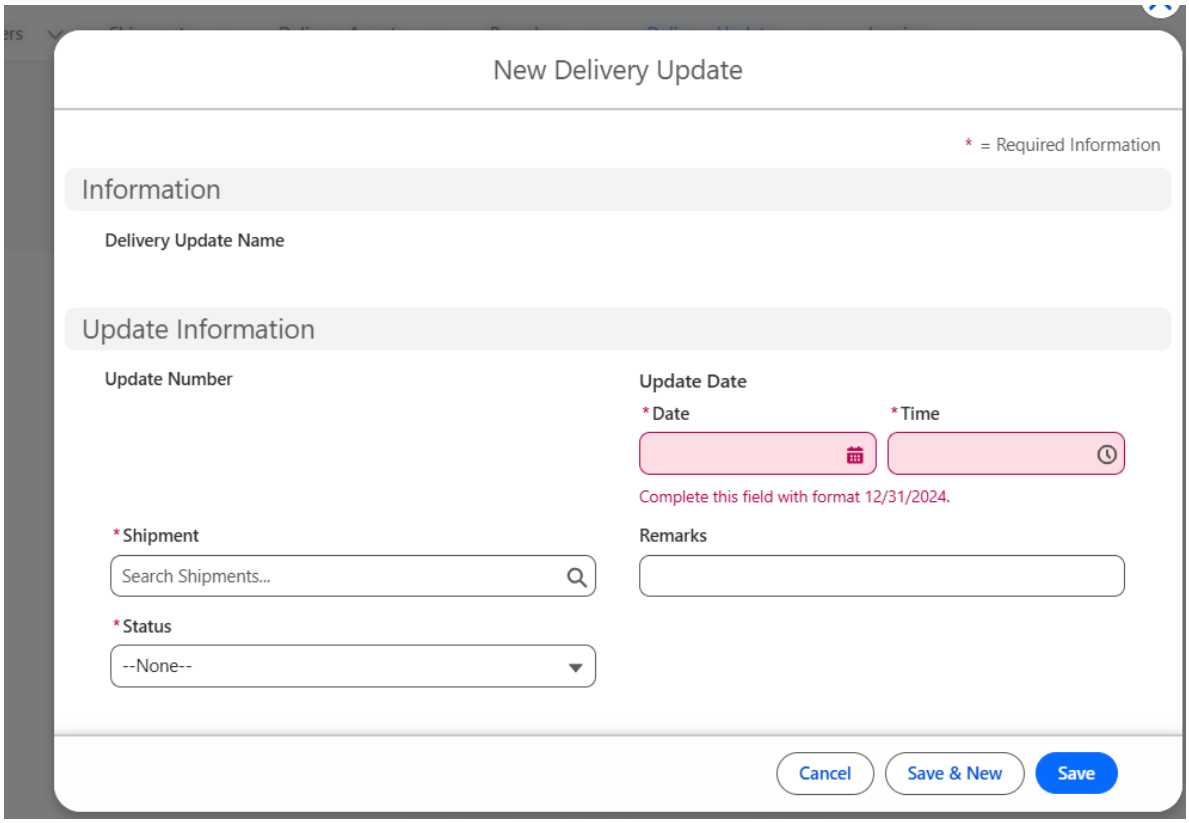


Figure 12 – Delivery Update used for shipment-status automation testing

6.6 Security and User Roles

The project uses role-based access control. The documented roles are Courier Admin, Branch Manager, Delivery Agent, and Customer Support. Profiles control permissions and roles control data visibility.

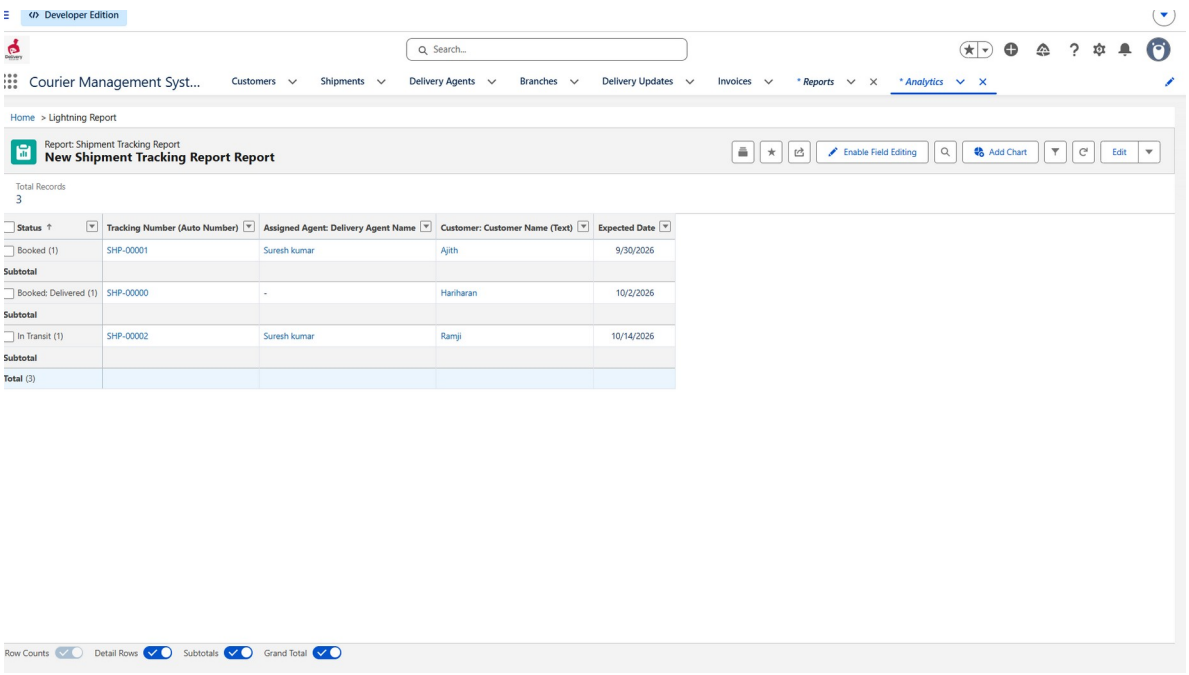
Role	Purpose
Courier Admin	Highest access for courier administration.
Branch Manager	Manages branch-level operations.
Delivery Agent	Access to assigned shipment operations.
Customer Support	Customer and shipment tracking support.

6.7 Reports and Dashboards

Three report types were configured: Shipment Tracking Report, Customer Summary Report, and Agent Performance Report.

Report	Configuration
Shipment Status Summary	Tracking Number, Customer, Status, Expected Date, Assigned Agent; grouped by Status; Price summarized.
Agent Performance Report	Agent Name, Branch, Number of Shipments, Status; grouped by Delivery Agent.
Revenue by Branch Report	Tracking Number, Customer, Price, Invoice Status; grouped by Branch and Month.

The Courier Management Dashboard contains the documented components: Shipment Status Overview (Donut Chart), Agent Performance (Bar Chart), and Revenue by Branch (Column Chart).



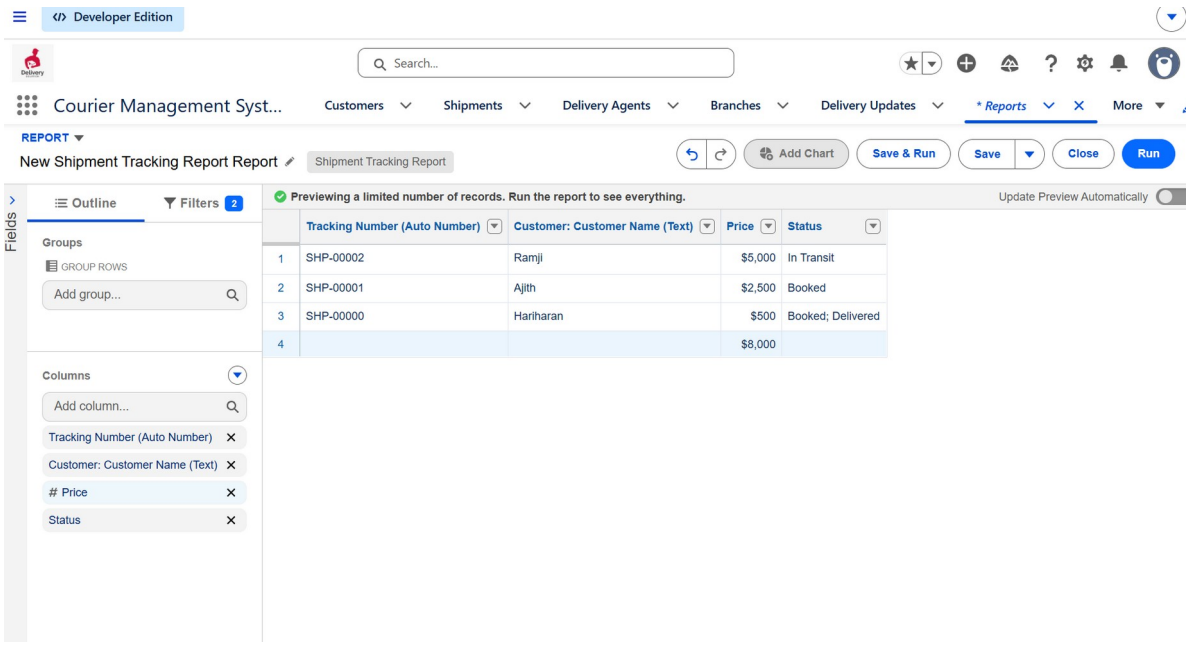


Figure 13 – Shipment Status Summary / Shipment Tracking report evidence

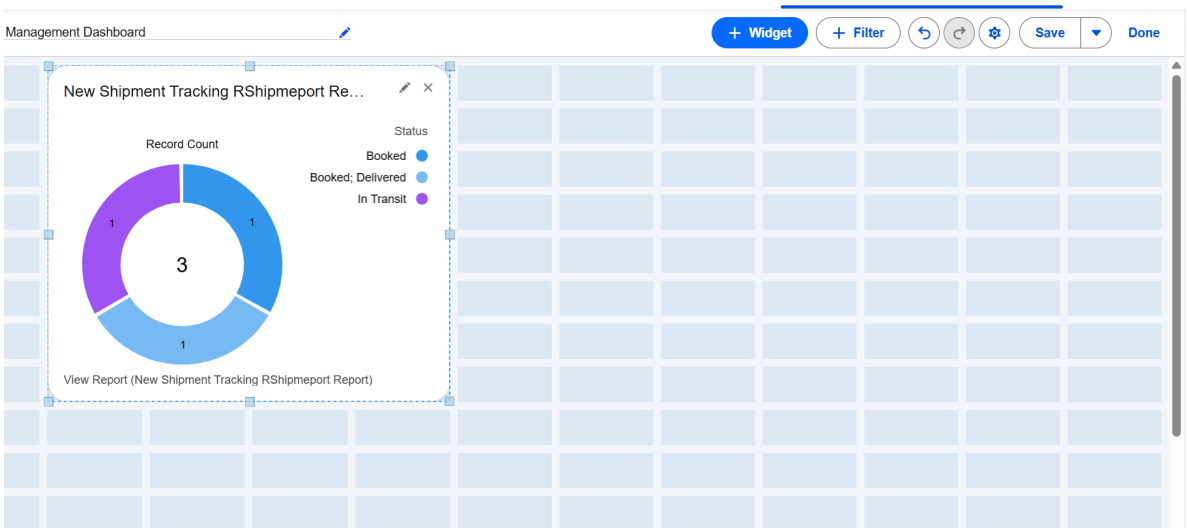


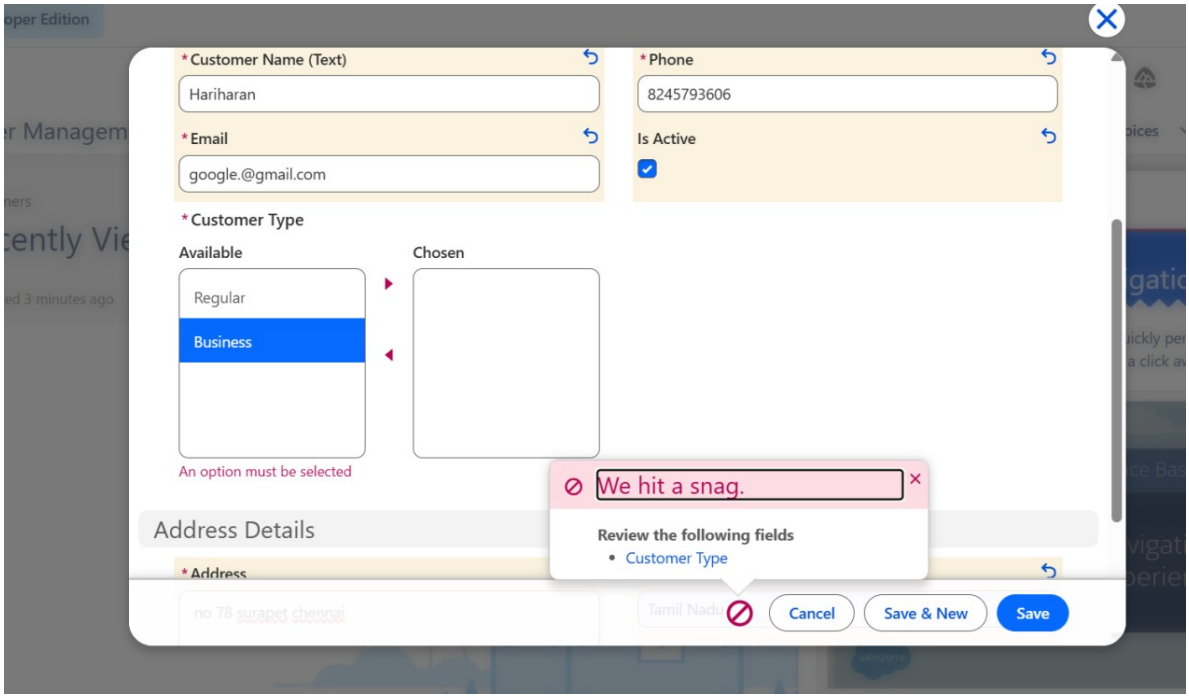
Figure 14 – Courier Management Dashboard

7. FUNCTIONAL AND PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to verify record creation, validation rule enforcement, delivery status updates, flow automation, invoice generation, and dashboard visualization. The documented testing approach validates the major functional areas of the system.

Test Scenario	Verification
Customer record creation	Create a valid customer record and verify that the record is saved correctly.
Customer validation	Verify email, phone number, and pincode validation rules.
Shipment validation	Verify weight, price, and expected delivery date validation rules.

Agent validation	Verify invalid email input is rejected.
Branch validation	Verify Branch Code cannot be blank.
Delivery Update validation	Verify remarks are required when status is Delayed.
Flow automation	Create a Delivery Update and verify that the related Shipment status is updated automatically.
Invoice generation	Create an invoice linked to a shipment and verify invoice details.
Reports	Run reports and verify displayed fields and grouping.
Dashboard visualization	Verify dashboard components and charts display operational data.



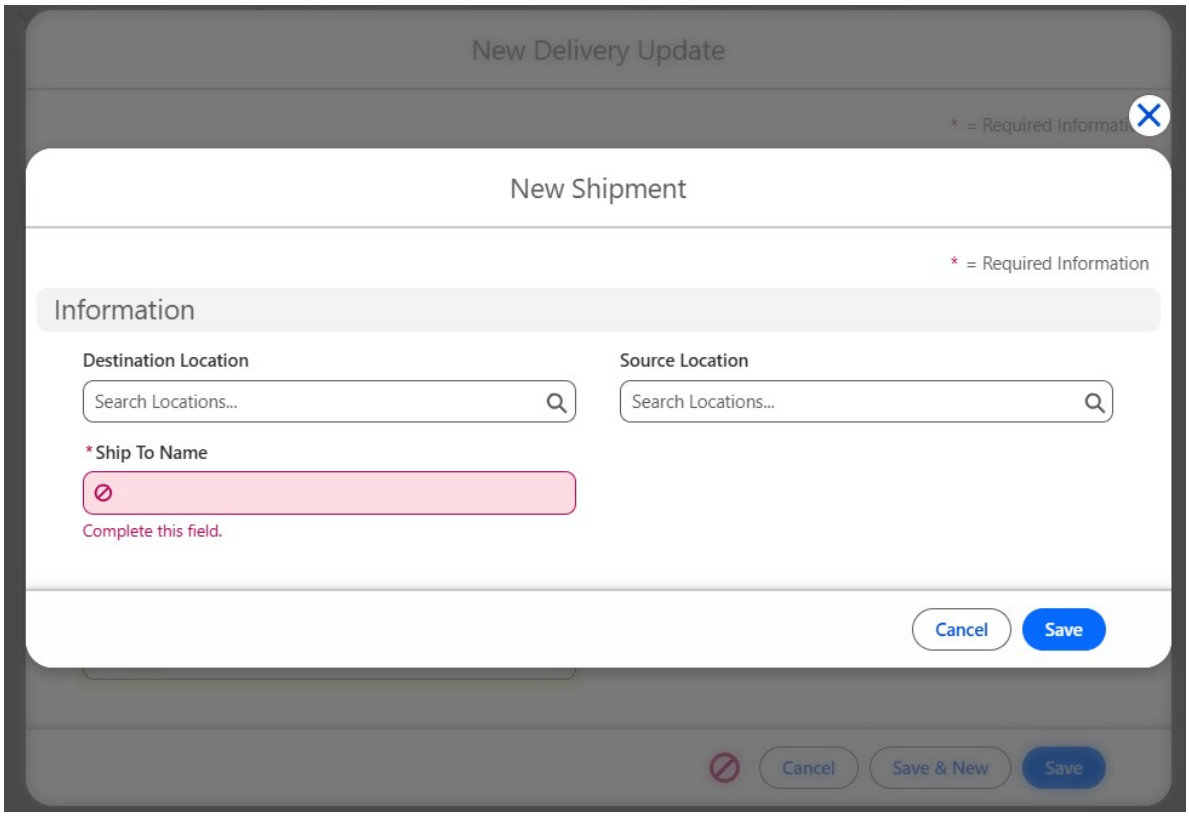


Figure 15 – Validation rule test results

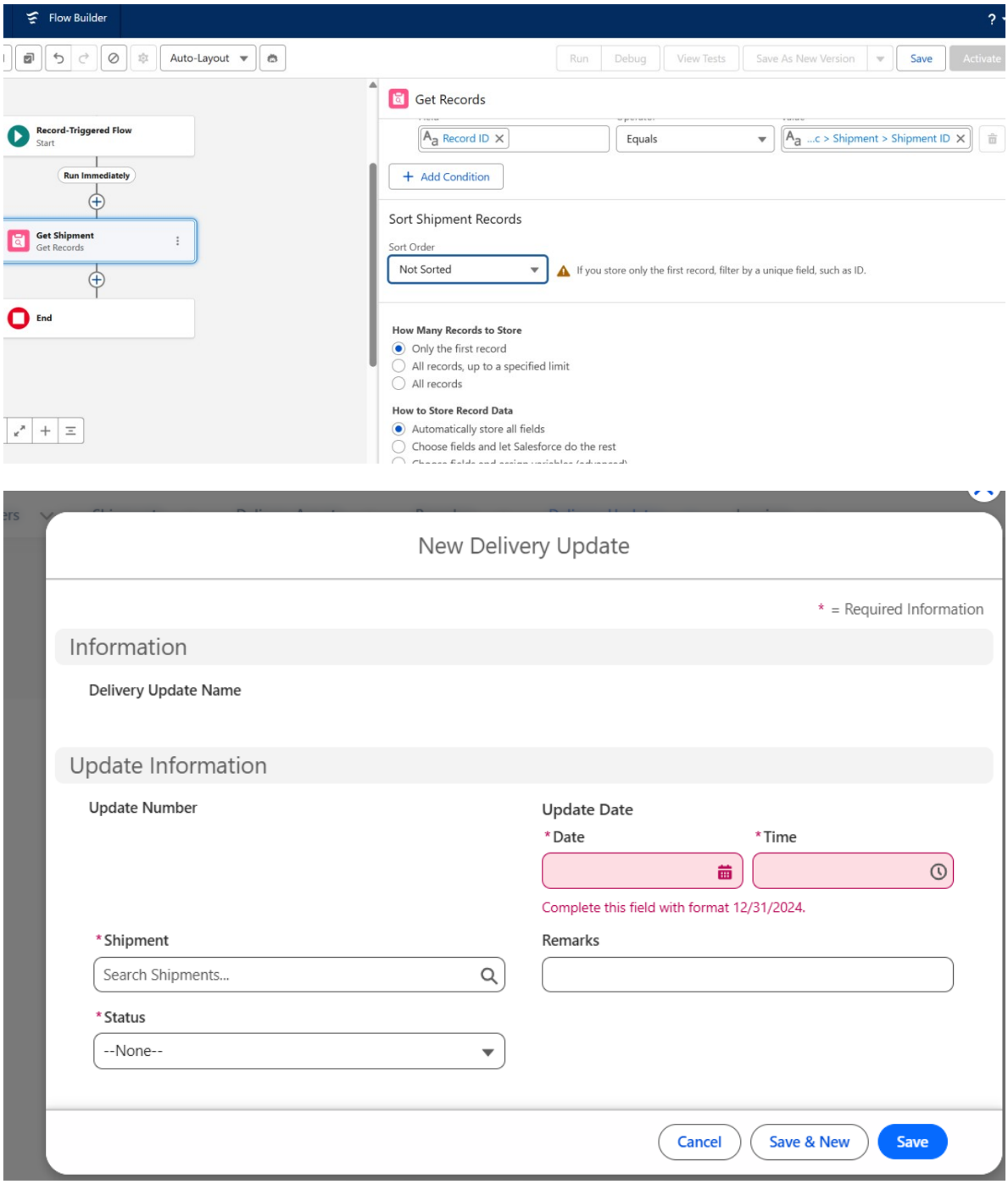


Figure 16 – Flow automation test evidence

8. ADVANTAGES AND DISADVANTAGES

8.1 Advantages

- Centralized data.
- Automated workflows.
- Real-time reporting.

8.2 Disadvantages

- Salesforce dependency.
- Requires initial configuration effort.

9. CONCLUSION

The Courier Management System successfully digitizes courier operations using Salesforce automation. The implementation brings together customer management, shipment tracking, delivery agent management, branch information, delivery updates, invoice management, validation, automation, reports, and dashboards in a centralized platform. The project is intended to improve efficiency, accuracy, operational visibility, and customer satisfaction.

10. FUTURE SCOPE

- Mobile tracking application.
- GPS integration.
- SMS notifications.
- AI-based delivery optimization.

11. APPENDIX – SCREENSHOT EVIDENCE

The project screenshots are embedded within the relevant implementation, testing, reporting, and dashboard sections to provide direct evidence of the Salesforce configuration and execution.